ALP-ttbar Toy MC v1.0–v1.2

Magnetic-Field Separation, Photon-Conversion Backgrounds, Vertex Resolution and a Preregistered Hit-Pattern Test for Displaced Light ALP $\rightarrow e^{+}e^{-}$

(companion to arXiv:2610.38932)

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About this compilation. This file combines two archived reports of a personal, numpy-only toy Monte Carlo written after reading J. Alimena, F. Blekman, C. Cazzaniga, X. Cid Vidal, J. Niedziela, L. Rygaard, R. Soto Rey, *Hunting for Long-Lived ALPs in Top Events via Displaced Dielectron Pairs at the LHC*, arXiv:2610.38932. The reports are reproduced unchanged. It is not affiliated with the authors or with the CMS Collaboration; only qualitative trends and ratios are meaningful.

Contents

Part	Version	Content	DOI
I (p. 2–5)	v1.1 (incl. v1.0)	B-field separation of collimated pairs; photon-conversion background; per-pair vertex resolution and tails	10.5281/zenodo.23833475
II (p. 6–8)	v1.2	Preregistered hit-pattern consistency test and lifetime scan (post-registration report)	10.5281/zenodo.23847880

Key results (toy, $c\tau$ = 1 mm, barrel)

- **v1.0:** the 3.8 T field raises the fraction of geometrically resolvable 17 MeV pairs from ~55% (straight lines) to ~97%.
- **v1.1:** light pairs vertex about as poorly as photon conversions; at 17 MeV the conversion-induced loss in reach ranges from ~2× to ~60×, driven mainly by resolution tails. At 100–200 MeV the result is stable.
- **v1.2:** a hit-pattern requirement was preregistered to reduce the worst 17 MeV case to 13–43×; the outcome was 55×, so this prediction was refuted. The v1.1 conclusion stands; the decisive quantity is the vertex-resolution tail for high-p_T collinear pairs, which requires detector-level simulation.

Links

Preregistration: github.com/wspaik5/ALP_ttbar_Toy_MC_v1.2_Preregistration_Hit_Pattern
Code (v1.2): github.com/wspaik5/ALP_ttbar_Toy_MC_v1.2_Postregistration_Hit_Pattern
Code (v1.0–v1.1): github.com/wspaik5/ALP_ttbar_Displaced_Dielectron_Toy_MC_arXiv2610.38932